

Which Summary Tells the Story?

AP Statistics · Unit 1 · Topic 1.7 · TEACHER KEY

CED TETHER

- ENDURING UNDERSTANDING: UNC-1 Graphical representations and statistics allow us to identify and represent key features of data.
- SKILLS:
- **Skill 2.C** — Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- LO UNC-1.I Calculate measures of center and position for quantitative data. [Skill 2.C]
- EK UNC-1.I.1 A statistic is a numerical summary of sample data.
- EK UNC-1.I.2 The mean is the sum of all the data values divided by the number of values. For a sample, the mean is denoted by $\bar{x}$: $\bar{x} = (1/n) \sum x_i$, where x_i represents the i^{th} data point in the sample and n represents the number of data values in the sample.
- EK UNC-1.I.3 The median of a data set is the middle value when data are ordered. When the number of data points is even, the median can take on any value between the two middle values. In AP Statistics, the most commonly used value for the median of a data set with an even number of values is the average of the two middle values.
- EK UNC-1.I.4 The first quartile, Q_1 , is the median of the half of the ordered data set from the minimum to the position of the median. The third quartile, Q_3 , is the median of the half of the ordered data set from the position of the median to the maximum. Q_1 and Q_3 form the boundaries for the middle 50% of values in an ordered data set.
- EK UNC-1.I.5 The p^{th} percentile is interpreted as the value that has $p\%$ of the data less than or equal to it.
- LO UNC-1.J Calculate measures of variability for quantitative data. [Skill 2.C]
- EK UNC-1.J.1 Three commonly used measures of variability (or spread) in a distribution are the range, interquartile range, and standard deviation.
- EK UNC-1.J.2 The range is defined as the difference between the maximum data value and the minimum data value. The interquartile range (IQR) is defined as the difference between the third and first quartiles: $Q_3 - Q_1$. Both the range and the interquartile range are possible ways of measuring variability of the distribution of a quantitative variable.
- EK UNC-1.J.3 Standard deviation is a way to measure variability of the distribution of a quantitative variable. For a sample, the standard deviation is denoted by s_x : $s_x = \sqrt{[(1/(n-1)) \sum (x_i - \bar{x})^2]}$. The square of the sample standard deviation, s^2 , is called the sample variance.
- EK UNC-1.J.4 Changing units of measurement affects the values of the calculated statistics.
- LO UNC-1.K Explain the selection of a particular measure of center and/or variability for describing a set of quantitative data. [Skill 4.B]
- EK UNC-1.K.1 There are many methods for determining outliers. Two methods frequently used in this course are:
 - i. An outlier is a value greater than $1.5 \times \text{IQR}$ above the third quartile or more than $1.5 \times \text{IQR}$ below the first quartile.
 - ii. An outlier is a value located 2 or more standard deviations above, or below, the mean.
- EK UNC-1.K.2 The mean, standard deviation, and range are considered nonresistant (or non-robust) because they are influenced by outliers. The median and IQR are considered resistant (or robust), because outliers do not greatly (if at all) affect their value.

Essential question: How can a statistic be calculated correctly but still be the wrong summary to report?

DOK-3 skill: 4.B · **FRQ pattern:** whose-computation-and-what-it-misleads-then-choose-the-summary

DOK rationale: Part (c) diagnoses two methods, selects a summary for purpose, and explains the rejected pair's consequence from the dotplot.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Where does the overall median go when you form the halves?
- Can correct arithmetic support a less representative summary?
- What changes if the 30-hour point moves farther right?
- What could 7.5 suggest without the dotplot?

[WARN] LOOK FOR

- Including the overall median in both halves.
- Treating either student as entirely correct.
- Choosing resistant summaries but retaining IQR 6.5.

SCORING PART (c) — E / P / I

Expected elements

- **quartile-method** (required) — Identifies Kai's IQR 8 and explains Rosa's inclusion of the overall median
- **corrected-pair** (required) — Chooses median 8 and IQR 8 hours
- **feature-and-resistance** (required) — Links isolated 30 or right skew to why mean/SD are more affected than the chosen pair

E	All three elements are correct and linked to this problem. Accept equivalent plain-language reasoning.
P	At least one element includes a correct evidence-to-reason link, but another is missing or incorrect.
I	No correct evidence-to-reason link; only a choice, copied numbers, or unsupported statements.

Common mistakes

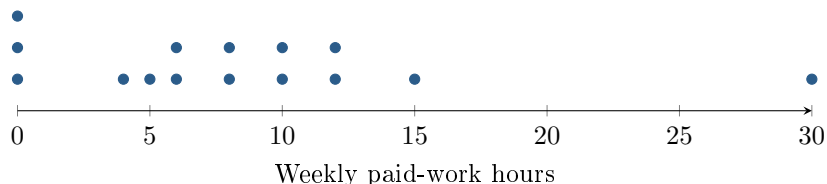
- Accepting one report wholesale instead of separating method from summary choice

[STAR] THE PROBLEM — annotated

Suppose a principal asks, “How much paid work do students do in a week?” Fifteen students report these weekly hours: 0, 0, 0, 4, 5, 6, 6, 8, 8, 10, 10, 12, 12, 15, 30.

Kai: “ $\bar{x} = 8.4$, $s_x \approx 7.5$, and $\text{IQR} = 8$. Report the mean and SD because they use every observation.”

Rosa: “Median = 8, $Q_1 = 4.5$, $Q_3 = 11$, and $\text{IQR} = 6.5$. Report the median and IQR because 30 looks unusual.”



FIRST TAKE — one sentence before you work the parts

Whose report would you send, Kai’s or Rosa’s? Cite one number or dotplot feature.

Key: Not graded. Preserve the split between computation and choice.

[BOOK] CENTER, SPREAD, AND OUTLIERS (keep on the page)

The median is the middle ordered value (average the two middle values for an even count). For this sheet, find Q_1 and Q_3 as the medians of the lower and upper halves, excluding the overall median when n is odd. $\text{IQR} = Q_3 - Q_1$. Outlier fences are $Q_1 - 1.5\text{IQR}$ and $Q_3 + 1.5\text{IQR}$. Mean and SD are nonresistant: extremes affect them more. Median and IQR are resistant. Use the distribution’s shape to choose a pair.

DOK 1 (a) State the median and maximum weekly paid-work hours.

Key: The median is 8 hours and the maximum is 30 hours.

DOK 2 (b) Using the printed quartile rule, find Q_1 , Q_3 , the IQR, and the upper 1.5-IQR fence. Is 30 beyond it? Use Kai’s mean and SD as given.

Key: Exclude the overall median: $Q_1 = 4$, $Q_3 = 12$, $\text{IQR} = 8$, and the upper fence is $12 + 1.5(8) = 24$ hours. The 30-hour value exceeds it. Kai’s supplied mean 8.4 and sample SD about 7.5 are correct; recalculating them is not required.

DOK 3 (c) Whose IQR follows the printed rule, and how did the other result arise? Choose a corrected summary pair for the principal. Use a dotplot feature to explain why that pair describes typical work hours better than the alternative.

Key: Kai’s IQR of 8 follows the printed rule. Rosa included the overall median in each half, giving quartiles 4.5 and 11 and IQR 6.5. Report median 8 and corrected IQR 8 hours: the isolated 30-hour value pulls the nonresistant mean and SD, especially the spread. The resistant pair describes the middle of the distribution without letting that one student’s hours dominate.